

Physical Chemistry (I)

Lecture 03

Kinetic Molecular Theory

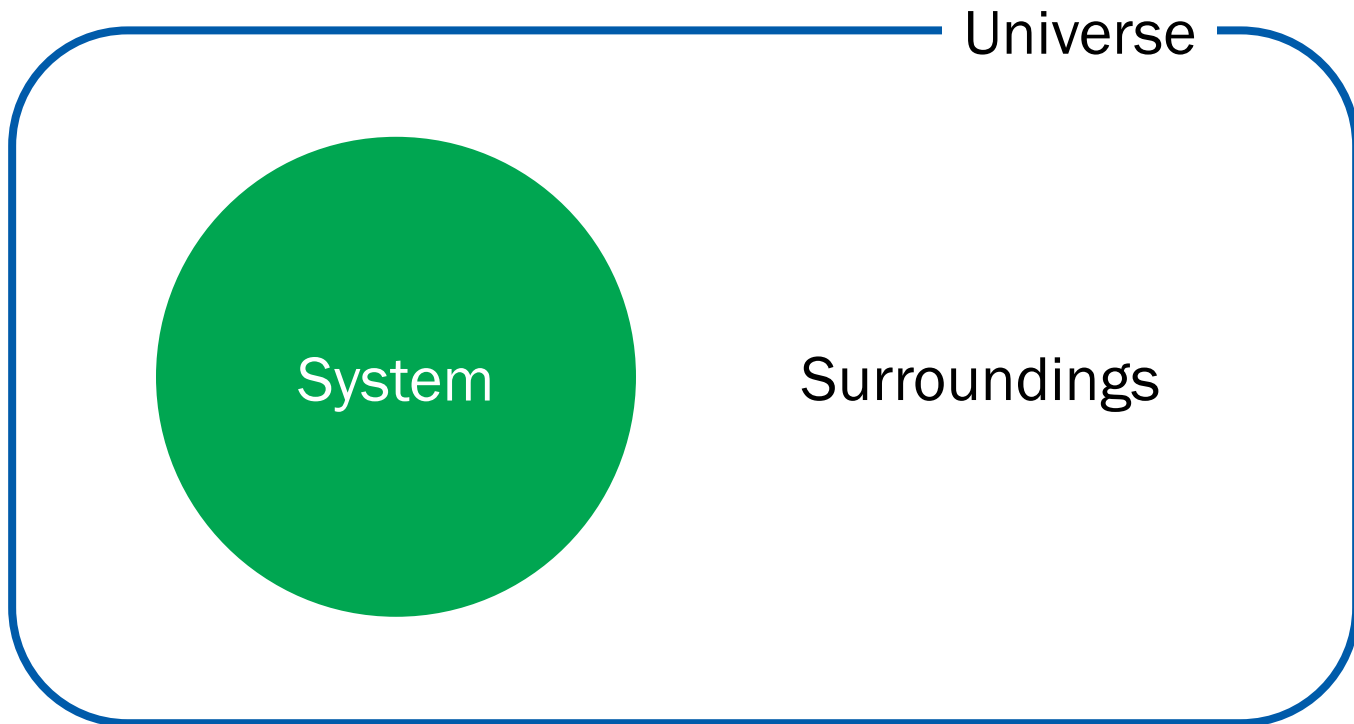


In the Last Class

- What is thermodynamics?
 1. Basic Concepts
 - System: properties, equilibrium
 - System state and state change
 2. The Properties of Gases
 - Why gases?
 - Perfect gas

Last Class

System and Surroundings



System Properties

- A system can have different **physical properties**
- A **single value** will be assigned for each property
- Useful classification
 - **Extensive properties:** $f(A) + f(B) = f(A + B)$
 - **Intensive properties:** $f(A) + f(B) \neq f(A + B)$

Equilibrium

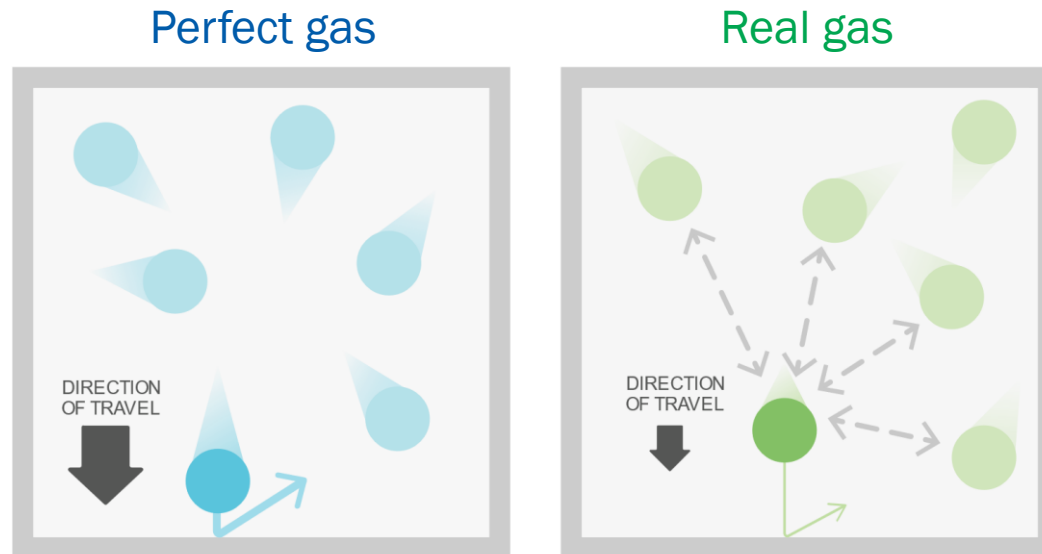
- An isolated system is **in equilibrium** when its properties remain **constant with time**
- Mechanical equilibrium: constant p and V
- Diffusive equilibrium: constant n
- Thermal equilibrium: constant T

State of a System

- The **state** of a system is defined by its **properties**
- For a pure substance: (n, V, p, T)
- **State change:** $(n_1, V_1, p_1, T_1) \rightarrow (n_2, V_2, p_2, T_2)$

Perfect Gas

- Perfect gas (ideal gas): no interactions
- Real gas: finite interactions between molecules



(Image: <https://cdn.kastatic.org/ka-perseus-images/4d1bf2d46543cc34f4948085bb9196f5dfb57515.svg>)

Equation of State

- The state of a pure gas is defined by (n, V, p, T)
- **Equation of state:** relationship of the 4 variables

$$p = f(T, V, n)$$

- Equation of state of a perfect gas: **perfect gas law**

$$p = nRT/V$$

↑
gas constant

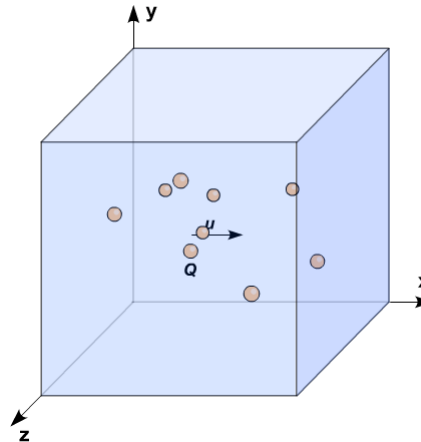
Kinetic Molecular Theory

Total energy = Σ (kinetic energies of molecules)

1. Molecules with mass m in ceaseless random motion
2. Molecules with negligible size
3. Elastic collisions: the total kinetic energy is conserved

Molecular Velocities

Each molecule has a different velocity

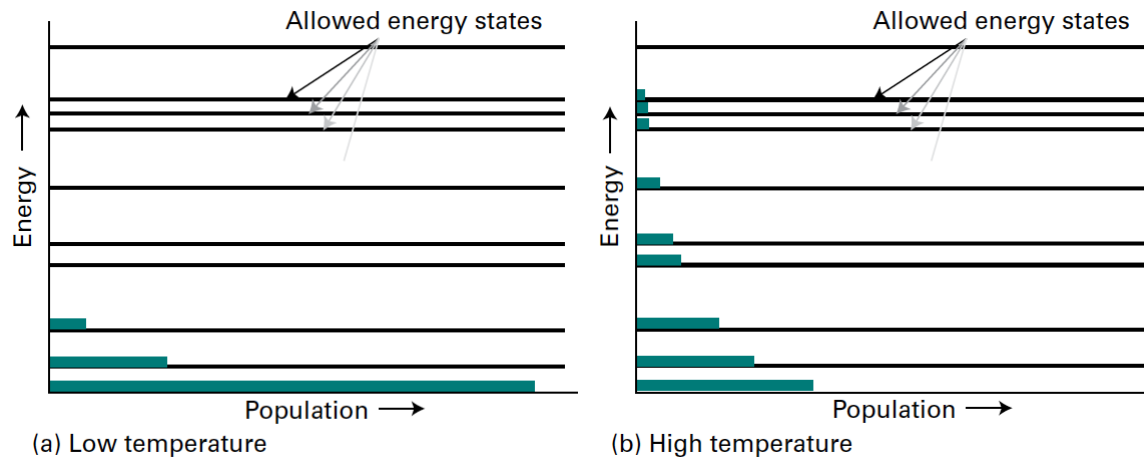


How can we determine the information about velocity?

Boltzmann Distribution

$$\text{Probability} \propto e^{-E/k_B T}$$

(at thermal equilibrium)



Probability: Example

Assume that we roll a fair dice. Probability is denoted as $p(k)$.

$$p(1) = p(2) = p(3) = p(4) = p(5) = p(6) = 1/6$$

- **Normalization:** $p(1) + p(2) + p(3) + p(4) + p(5) + p(6) = 1$

$$\sum_{k=1}^6 p(k) = 1$$

- **Average:** $\langle k \rangle = (1 + 2 + 3 + 4 + 5 + 6)/6 = 3.5$

$$\langle k \rangle = 1 \cdot p(1) + 2 \cdot p(2) + 3 \cdot p(3) + 4 \cdot p(4) + 5 \cdot p(5) + 6 \cdot p(6)$$

$$= \sum_{k=1}^6 k \cdot p(k)$$

Probability: Example

Now, we play a dice game with the following rule for a gain:

$$f(1) = \cancel{\text{¥}}10,000, f(2) = \cancel{\text{¥}}5,000, f(3) = f(4) = f(5) = f(6) = 0$$

- Expectation for the gain:

$$\langle f \rangle = \cancel{\text{¥}}10,000 \times (1/6) + \cancel{\text{¥}}5,000 \times (1/6) + 0 \times (4/6) = \cancel{\text{¥}}2,500$$

$$\begin{aligned}\langle f \rangle &= f(1)p(1) + f(2)p(2) + f(3)p(3) \\ &\quad + f(4)p(4) + f(5)p(5) + f(6)p(6) \\ &= \sum_{k=1}^6 f(k)p(k)\end{aligned}$$

Summary

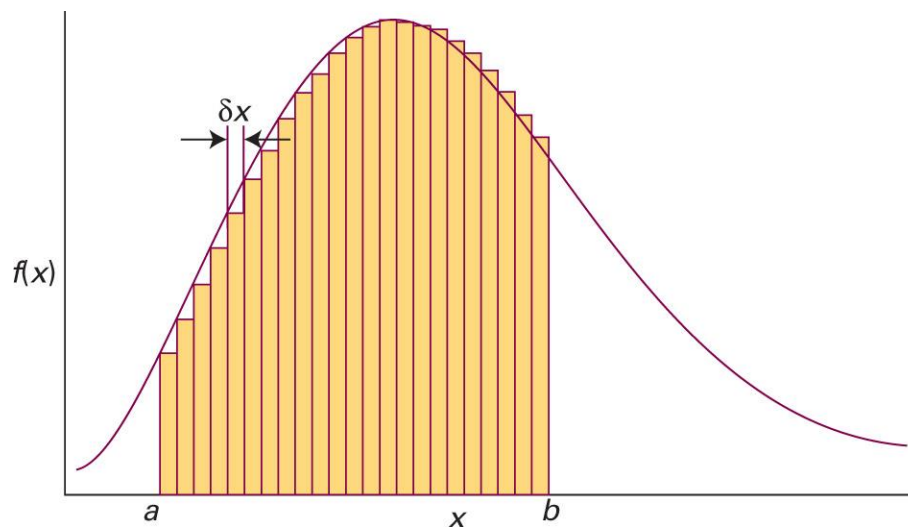
	Discrete
Variable	$k = 1, 2, \dots, N$
Probability	$p(k)$
Normalization	$\sum_{k=1}^N p(k) = 1$
Average	$\langle k \rangle = \sum_{k=1}^N kp(k)$
Expectation	$\langle f \rangle = \sum_{k=1}^N f(k)p(k)$

From Discrete to Continuous

	Discrete	Continuous
Variable	$k = 1, 2, \dots, N$	x : real in $[x_1, x_2]$
Probability	$p(k)$	
Normalization	$\sum_{k=1}^N p(k) = 1$	
Average	$\langle k \rangle = \sum_{k=1}^N kp(k)$	
Expectation	$\langle f \rangle = \sum_{k=1}^N f(k)p(k)$	

Integration

“The areas under curves”



$$\lim_{\delta x \rightarrow 0} \sum_i f(x_i) \delta x = \int_a^b f(x) dx$$

(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., the chemist's toolkit 4, Sketch 1)

Common Integrals (Part A)

Indefinite integral*	Constraint	Definite integral
Algebraic functions		
A.1 $\int x^n dx = \frac{1}{n+1} x^{n+1} + c$	$n \neq -1$	$\int_a^b x^n dx = \frac{1}{n+1} (b^{n+1} - a^{n+1})$
A.2 $\int \frac{1}{x} dx = \ln x + c$		$\int_a^b \frac{1}{x} dx = \ln \frac{b}{a}$
A.3 $\int \frac{1}{(A-x)(B-x)} dx = \frac{1}{B-A} \ln \frac{B-x}{A-x} + c$	$A \neq B$	$\int_a^b \frac{1}{(A-x)(B-x)} dx = \frac{1}{B-A} \ln \frac{(B-b)(A-a)}{(A-b)(B-a)}$
Exponential functions		
E.1 $\int e^{-kx} dx = -\frac{1}{k} e^{-kx} + c$		$\int_a^b e^{-kx} dx = \frac{1}{k} (e^{-ka} - e^{-kb})$
E.2 $\int x e^{-kx} dx = -\frac{1}{k^2} e^{-kx} - \frac{x}{k} e^{-kx} + c$		$\int_a^b x e^{-kx} dx = -\frac{1}{k^2} (e^{-kb} - e^{-ka}) - \frac{1}{k} (b e^{-kb} - a e^{-ka})$
E.3	$n \geq 0, k > 0$	$\int_0^\infty x^n e^{-kx} dx = \frac{n!}{k^{n+1}} \quad n! = n(n-1) \dots 1; 0! = 1$
E.4		$\int_0^\infty \frac{x^4 e^x}{(e^x - 1)^2} dx = \frac{4\pi^4}{15}$
Gaussian functions		
G.1		$\int_0^\infty e^{-kx^2} dx = \frac{1}{2} \left(\frac{\pi}{k} \right)^{1/2}$
G.2 $\int x e^{-kx^2} dx = -\frac{1}{2k} e^{-kx^2} + c$		$\int_0^\infty x e^{-kx^2} dx = \frac{1}{2k}$
G.3		$\int_0^\infty x^2 e^{-kx^2} dx = \frac{1}{4} \left(\frac{\pi}{k^3} \right)^{1/2}$
G.4 $\int x^3 e^{-kx^2} dx = -\frac{1}{2k^2} e^{-kx^2} - \frac{x^2}{2k} e^{-kx^2} + c$	$k > 0$	$\int_0^\infty x^3 e^{-kx^2} dx = \frac{1}{2k^2}$
G.5	$k > 0$	$\int_0^\infty x^4 e^{-kx^2} dx = \frac{3}{8k^2} \left(\frac{\pi}{k} \right)^{1/2}$

Probability Distributions

	Discrete	Continuous
Variable	$k = 1, 2, \dots, N$	x : real in $[x_1, x_2]$
Probability	$p(k)$	$p(x)dx$ probability density ↘
Normalization	$\sum_{k=1}^N p(k) = 1$	$\int_{x_1}^{x_2} p(x) dx = 1$
Average	$\langle k \rangle = \sum_{k=1}^N kp(k)$	$\langle x \rangle = \int_{x_1}^{x_2} xp(x) dx$
Expectation	$\langle f \rangle = \sum_{k=1}^N f(k)p(k)$	$\langle f \rangle = \int_{x_1}^{x_2} f(x)p(x) dx$

Properties of Probabilities

- $\langle kf(x) \rangle = k\langle f(x) \rangle$

$$\langle kf(x) \rangle = \int_{x_1}^{x_2} [kf(x)]p(x) dx = k \int_{x_1}^{x_2} f(x)p(x) dx = k\langle f(x) \rangle$$

- $\langle f(x) + g(x) \rangle = \langle f(x) \rangle + \langle g(x) \rangle$

$$\begin{aligned}\langle f(x) + g(x) \rangle &= \int_{x_1}^{x_2} [f(x) + g(x)]p(x) dx \\ &= \int_{x_1}^{x_2} f(x)p(x) dx + \int_{x_1}^{x_2} g(x)p(x) dx = \langle f(x) \rangle + \langle g(x) \rangle\end{aligned}$$

Boltzmann Distribution

- Probability density for energy E

$$f(E) \propto e^{-E/k_{\text{B}}T}$$

$$f(E) = K e^{-E/k_{\text{B}}T}$$

- $f(E)dE$: the probability that a molecule has an energy in the range $[E, E + dE]$

Probability Density for Velocity

$$f(E) = K e^{-E/k_B T}$$

Since the energy of a molecule = its kinetic energy,

$$E = \frac{1}{2} m v^2 = \frac{1}{2} m v_x^2 + \frac{1}{2} m v_y^2 + \frac{1}{2} m v_z^2$$

Probability density for velocity (v_x, v_y, v_z) :

$$f(v_x, v_y, v_z) = K e^{-m v_x^2 / 2 k_B T} e^{-m v_y^2 / 2 k_B T} e^{-m v_z^2 / 2 k_B T}$$

Separation of Variables

$$f(v_x, v_y, v_z) = K e^{-mv_x^2/2k_B T} e^{-mv_y^2/2k_B T} e^{-mv_z^2/2k_B T}$$

Let $f(v_x, v_y, v_z) = f(v_x)f(v_y)f(v_z)$, where

$$f(v_x) = K_x e^{-mv_x^2/2k_B T}$$

$$f(v_y) = K_y e^{-mv_y^2/2k_B T}$$

$$f(v_z) = K_z e^{-mv_z^2/2k_B T}$$

such that

$$K = K_x K_y K_z$$

Determination of K 's

$$f(v_x) = K_x e^{-mv_x^2/2k_B T}$$

Use the normalization condition.

Since the range of v_x is $(-\infty, \infty)$,

$$\int_{-\infty}^{\infty} f(v_x) dv_x = 1$$

$$\int_{-\infty}^{\infty} K_x e^{-mv_x^2/2k_B T} dv_x = 1$$

Determination of K 's

$$\int_{-\infty}^{\infty} f(v_x) dv_x = 1$$

$$\int_{-\infty}^{\infty} K_x e^{-mv_x^2/2k_B T} dv_x = 1$$

$$K_x \int_{-\infty}^{\infty} e^{-mv_x^2/2k_B T} dv_x = 1$$

$$K_x \left(\frac{2\pi k_B T}{m} \right)^{1/2} = 1$$

$$K_x = \left(\frac{m}{2\pi k_B T} \right)^{1/2}$$

Normalized Expression

$$K_x = \left(\frac{m}{2\pi k_B T} \right)^{1/2} = K_y = K_z$$

$$\begin{aligned} f(v_x, v_y, v_z) &= K_x K_y K_z e^{-mv_x^2/2k_B T} e^{-mv_y^2/2k_B T} e^{-mv_z^2/2k_B T} \\ &= \left(\frac{m}{2\pi k_B T} \right)^{3/2} e^{-mv_x^2/2k_B T} e^{-mv_y^2/2k_B T} e^{-mv_z^2/2k_B T} \\ &= \left(\frac{m}{2\pi k_B T} \right)^{3/2} e^{-m(v_x^2 + v_y^2 + v_z^2)/2k_B T} \end{aligned}$$

Velocity to Speed

- $f(v_x, v_y, v_z) dv_x dv_y dv_z$:

Probability that a molecule has a velocity in the range

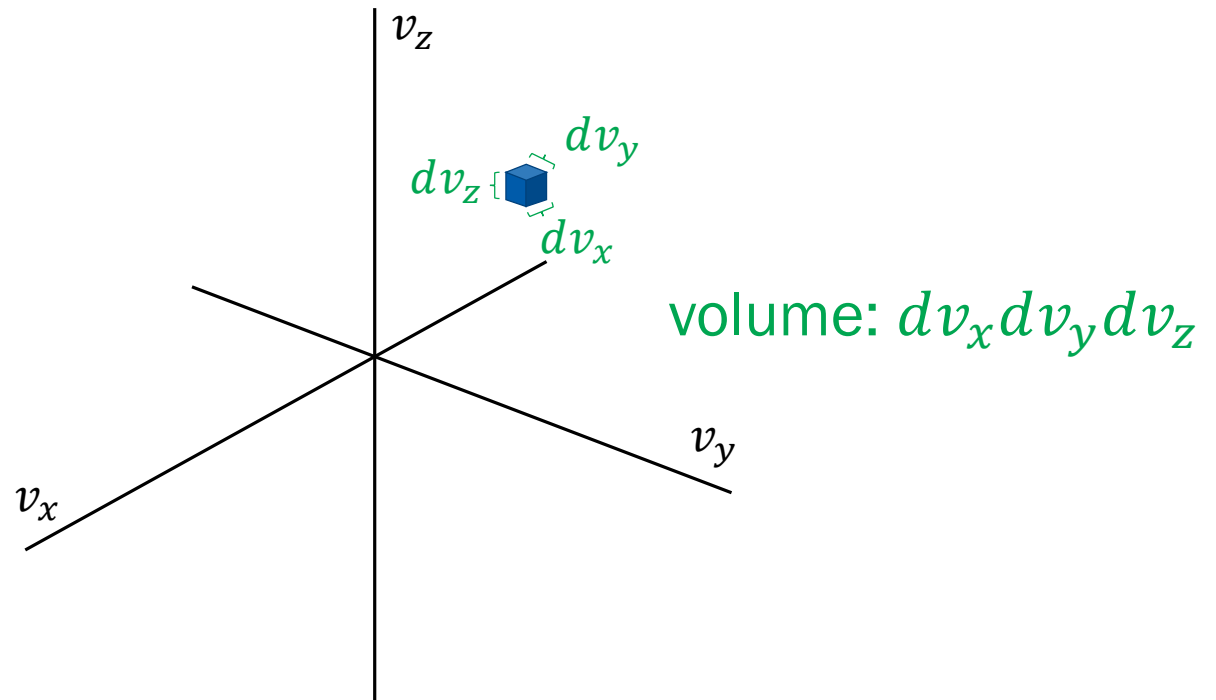
$$[v_x, v_x + dv_x], [v_y, v_y + dv_y], [v_z, v_z + dv_z]$$

- $f(v)dv$:

Probability that a molecule has a speed in the range $[v, v + dv]$

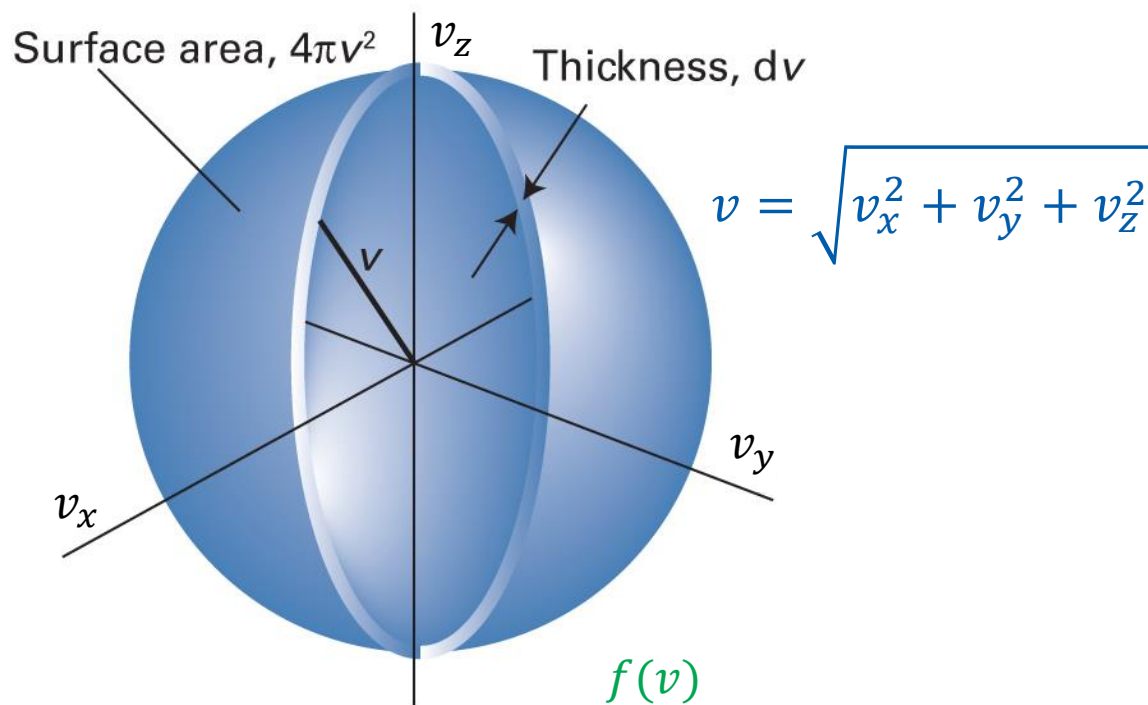
Can we obtain $f(v)dv$ from $f(v_x, v_y, v_z) dv_x dv_y dv_z$?

Velocity Space



$$f(v_x, v_y, v_z) dv_x dv_y dv_z = \text{function} \times \text{volume}$$

Velocity Space



$$\text{function} \times \text{volume} = \overbrace{f(v_x, v_y, v_z)}^{f(v)} \times 4\pi v^2 dv$$

Maxwell-Boltzmann Distribution

$$f(v) = f(v_x, v_y, v_z) 4\pi v^2$$

$$f(v) = 4\pi \left(\frac{m}{2\pi k_B T} \right)^{3/2} v^2 e^{-m(v_x^2 + v_y^2 + v_z^2)/2k_B T}$$

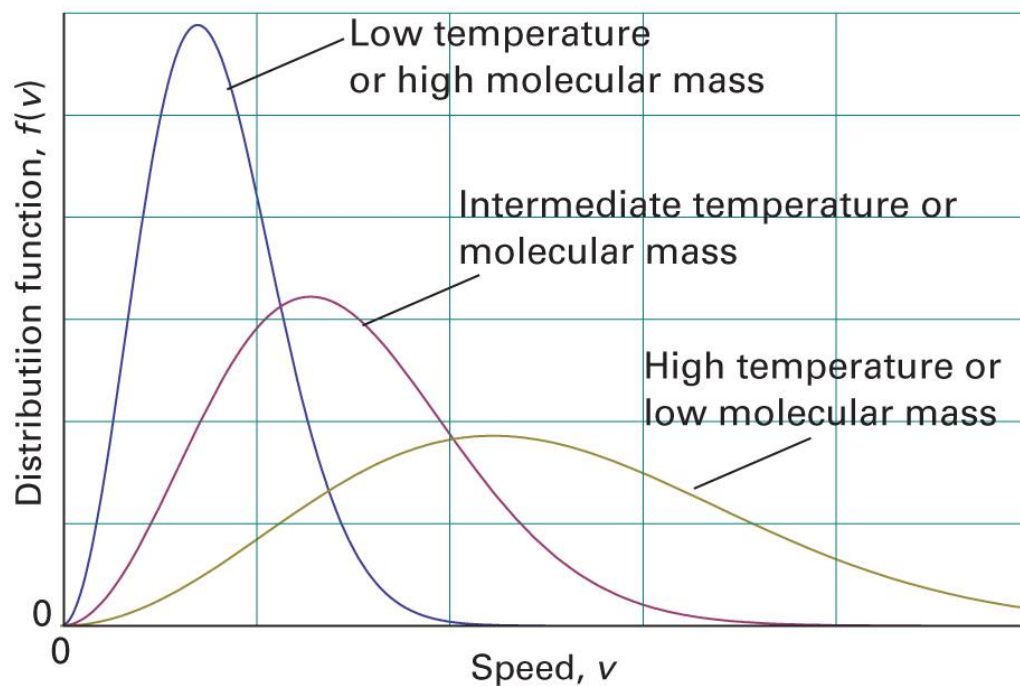
$$f(v) = 4\pi \left(\frac{m}{2\pi k_B T} \right)^{3/2} v^2 e^{-mv^2/2k_B T}$$

$$f(v) = 4\pi \left(\frac{M}{2\pi R T} \right)^{3/2} v^2 e^{-Mv^2/2RT}$$

$R = k_B N_A$
 $M = m N_A$

Interpretations

$$f(v) = 4\pi \left(\frac{M}{2\pi RT} \right)^{3/2} v^2 e^{-Mv^2/2RT}$$



(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figure 1B.4)

Characteristic Speeds

1. Mean speed $\langle v \rangle = v_{\text{mean}}$

$$\langle v \rangle = \int_0^{\infty} v f(v) dv$$

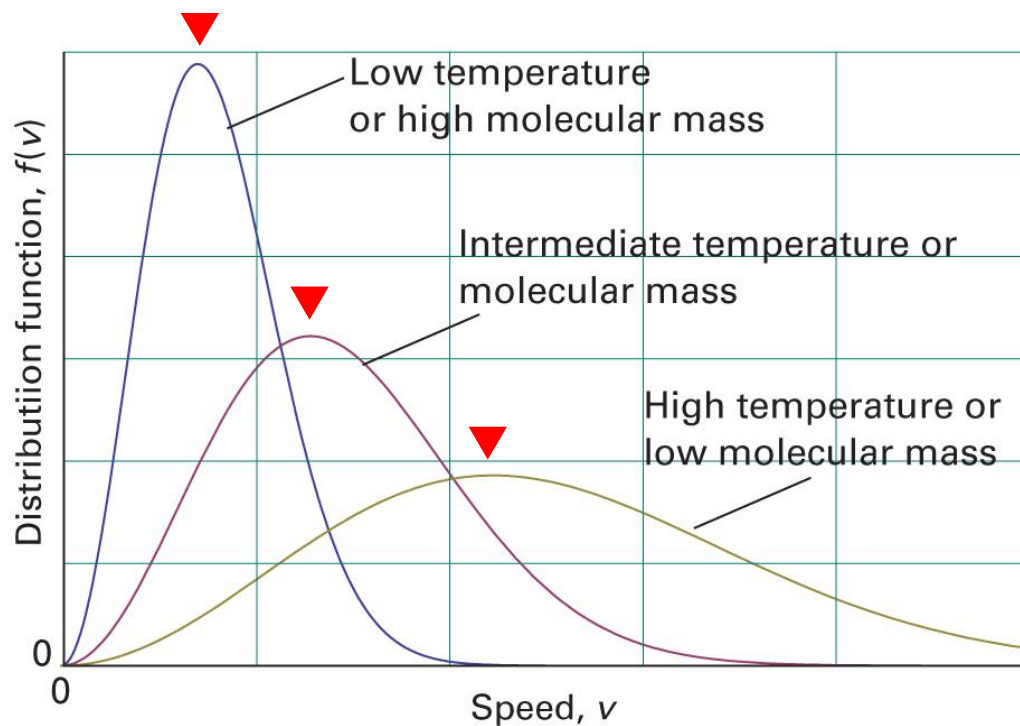
2. Root-mean-square speed v_{rms}

$$v_{\text{rms}} = \sqrt{\langle v^2 \rangle}$$

$$\langle v^2 \rangle = \int_0^{\infty} v^2 f(v) dv$$

Characteristic Speeds

3. Most probable speed v_{mp}



$$\frac{df(v)}{dv} = 0$$

(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figure 1B.4)

Characteristic Speeds: KMT

1. Mean speed

$$\langle v \rangle = \left(\frac{8RT}{\pi M} \right)^{1/2}$$

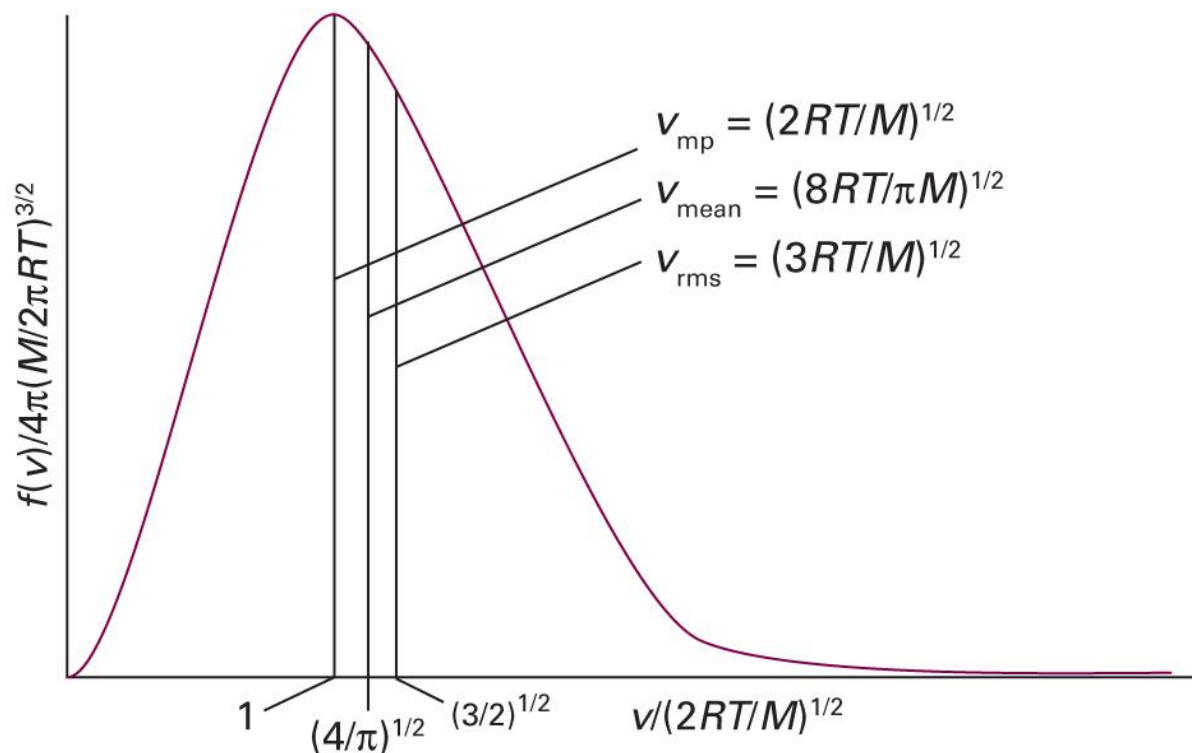
2. Root-mean-square speed

$$v_{\text{rms}} = \left(\frac{3RT}{M} \right)^{1/2}$$

3. Most probable speed

$$v_{\text{mp}} = \left(\frac{2RT}{M} \right)^{1/2}$$

Characteristic Speeds: KMT



(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figure 1B.6)

Example 1B.1

Calculate v_{rms} and the mean speed, $\langle v \rangle$, of N_2 molecules at 25°C .

For N_2 , $M = 28.02 \text{ g mol}^{-1} = 0.02802 \text{ kg mol}^{-1}$

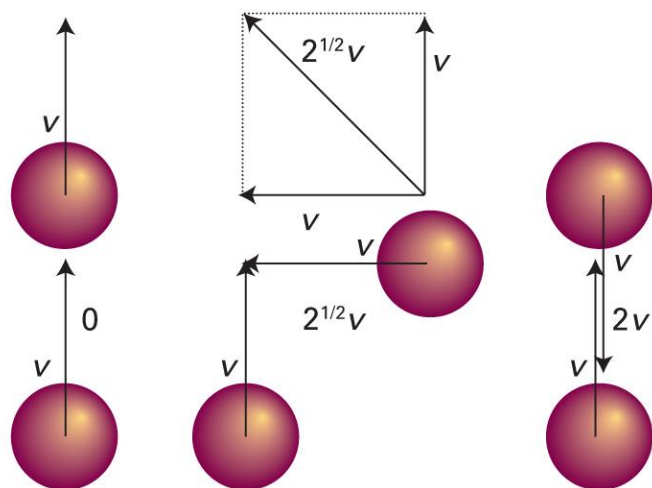
$T = (273 + 25) \text{ K} = 298 \text{ K}$

$$v_{\text{rms}} = \left(\frac{3RT}{M} \right)^{1/2} = \left\{ \frac{3 \times (8.3145 \text{ J K}^{-1} \text{ mol}^{-1}) \times (298 \text{ K})}{0.02802 \text{ kg mol}^{-1}} \right\}^{1/2} = 515 \text{ m s}^{-1}$$

$$\langle v \rangle = \left(\frac{8RT}{\pi M} \right)^{1/2} = \left\{ \frac{8 \times (8.3145 \text{ J K}^{-1} \text{ mol}^{-1}) \times (298 \text{ K})}{3.1416 \times 0.02802 \text{ kg mol}^{-1}} \right\}^{1/2} = 475 \text{ m s}^{-1}$$

Mean Relative Speed

“Mean speed with which a molecule approaches another”



$$v_{\text{rel}} = 2^{1/2} \langle v \rangle = \left(\frac{16RT}{\pi M} \right)^{1/2}$$

(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figure 1B.7)

Mean Relative Speed

For two molecules of different types (A and B),

$$v_{\text{rel}} = \left(\frac{8k_{\text{B}}T}{\pi\mu} \right)^{1/2}$$

where reduced mass $\mu = m_{\text{A}}m_{\text{B}}/(m_{\text{A}} + m_{\text{B}})$

If $m_{\text{A}} = m_{\text{B}}$,

$$\mu = m_{\text{A}}^2/(2m_{\text{A}}) = m_{\text{A}}/2$$

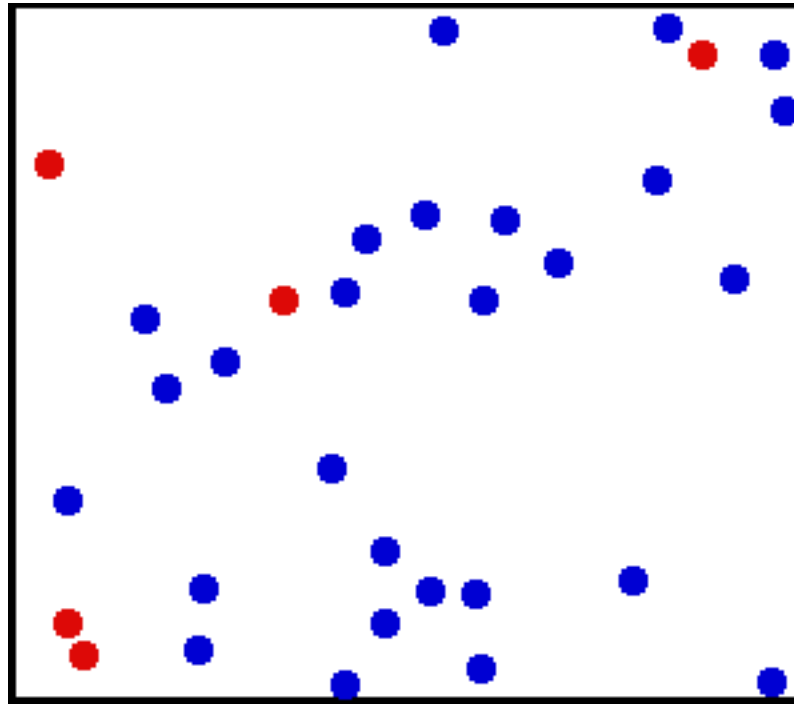
$$v_{\text{rel}} = \left(\frac{16k_{\text{B}}T}{\pi m} \right)^{1/2} = \left(\frac{16RT}{\pi M} \right)^{1/2}$$

Brief Illustration 1B.1

The mean speed of N_2 molecules at 25°C is 475 m s^{-1} . What is their relative mean speed?

$$v_{\text{rel}} = 2^{1/2} \langle v \rangle = \sqrt{2} \times (475\text{ m s}^{-1}) = 671\text{ m s}^{-1}$$

Application: Collisions



(Image: https://commons.wikimedia.org/wiki/File:Translational_motion.gif)

Consequences of Collisions

- Molecule-molecule collisions

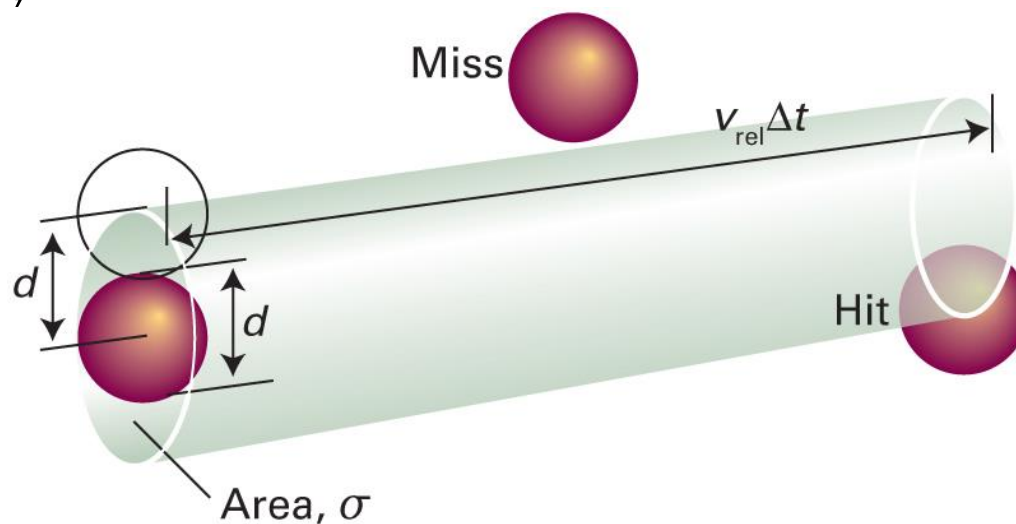
Molecular origin of chemical reactions

- Molecule-wall collisions

Molecular origin of pressure

Molecule-Molecule Collisions

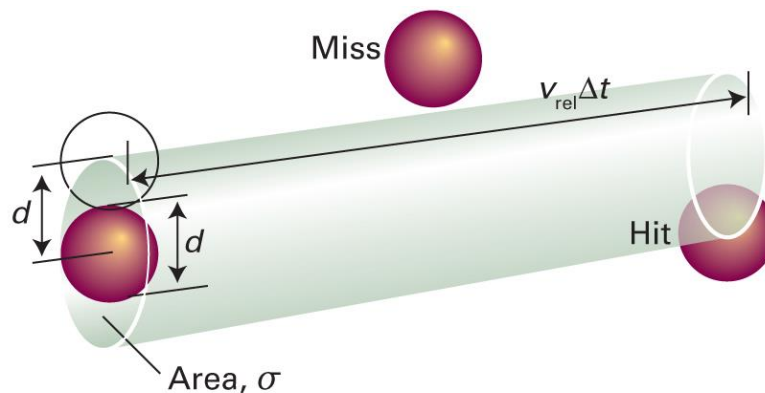
For time Δt ,



d = collision diameter

$\sigma = \pi d^2$ = cross-sectional area

Collision Frequency



- Volume of the collision tube = $\sigma \times v_{\text{rel}} \Delta t$
- Number of molecules in the tube = volume $\times$ density

$$= (\sigma v_{\text{rel}} \Delta t) \times (nN_A/V)$$
- Collision frequency $z = \sigma v_{\text{rel}} nN_A/V$

T*- and *p*-dependences of *z

$$z = \frac{\sigma v_{\text{rel}} n N_{\text{A}}}{V} = \sigma v_{\text{rel}} N_{\text{A}} \frac{n}{V}$$

Since $v_{\text{rel}} = (16RT/\pi M)^{1/2}$,

$$z = \sigma N_{\text{A}} \frac{n}{V} \left(\frac{16RT}{\pi M} \right)^{1/2} \propto \sqrt{T} \quad (\text{at constant } V)$$

Since $pV = nRT$,

$$z = \sigma v_{\text{rel}} N_{\text{A}} \frac{p}{RT} = \sigma v_{\text{rel}} \frac{p}{k_{\text{B}}T} \propto p \quad (\text{at constant } T)$$

Brief Illustration 1B.2

What is the collision frequency of an N_2 molecule in a sample at 1.00 atm and 25°C ? ($\sigma = 0.45 \text{ nm}^2$)

From Brief Illustration 1B.1, $v_{\text{rel}} = 671 \text{ m s}^{-1}$.

$$z = \frac{\sigma v_{\text{rel}} p}{k_{\text{B}} T} = \frac{(0.45 \times 10^{-18} \text{ m}^2) \times (671 \text{ m s}^{-1}) \times (1.01 \times 10^5 \text{ Pa})}{(1.381 \times 10^{-23} \text{ J K}^{-1}) \times (298 \text{ K})}$$
$$z = 7.4 \times 10^9 \text{ s}^{-1}$$

Mean Free Path, λ

“Average distance a molecule travels between collisions”

$$\lambda = \text{speed} \times \text{time} = v_{\text{rel}} \times (1/z) = v_{\text{rel}}/z$$

Hence,

$$\lambda = \frac{k_B T}{\sigma p}$$

Brief Illustration 1B.3

What is the mean free path of an N_2 molecule in a sample at 1.00 atm and 25 °C?

From Brief Illustration 1B.1, $v_{\text{rel}} = 671 \text{ m s}^{-1}$.

From Brief Illustration 1B.2, $z = 7.4 \times 10^9 \text{ s}^{-1}$.

$$\lambda = \frac{v_{\text{rel}}}{z} = \frac{671 \text{ m s}^{-1}}{7.4 \times 10^9 \text{ s}^{-1}} = 9.1 \times 10^{-8} \text{ m}$$

Prelude to Chemical Kinetics

$$z = \sigma v_{\text{rel}} N_A \frac{n}{V}$$

Suppose that we have two types of molecules, A and B.

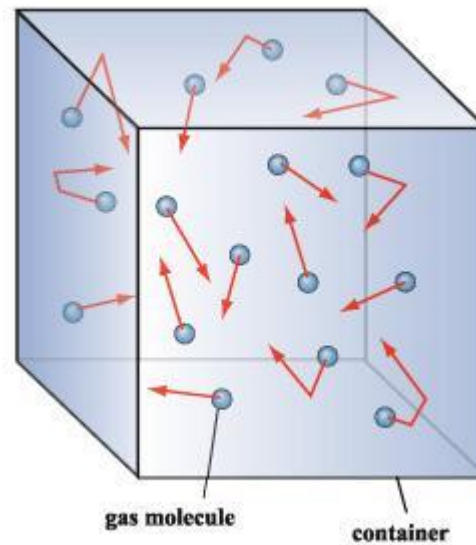
The collision rate of one A molecule with B molecules is

$$z = \sigma v_{\text{rel}} N_A \frac{n_B}{V}$$

Hence, the collision density $Z_{AB} = z(N_A n_A / V)$

$$Z_{AB} = \sigma v_{\text{rel}} N_A^2 \frac{n_A}{V} \frac{n_B}{V} = \sigma \left(\frac{8k_B T}{\pi \mu} \right)^{1/2} N_A^2 [A][B] \propto [A][B]$$

Molecular Origin of Pressure

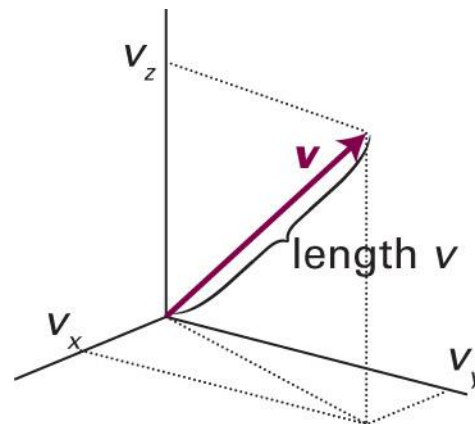


Position and Velocity

Each particle with mass m has its own

- Position $\vec{r} = (x, y, z)$
- Velocity $\vec{v} = (v_x, v_y, v_z)$

$$\vec{v} = \frac{d\vec{r}}{dt} = \left(\frac{dx}{dt}, \frac{dy}{dt}, \frac{dz}{dt} \right)$$



We can define linear momentum $\vec{p} = m\vec{v}$

Newton's Second Law of Motion

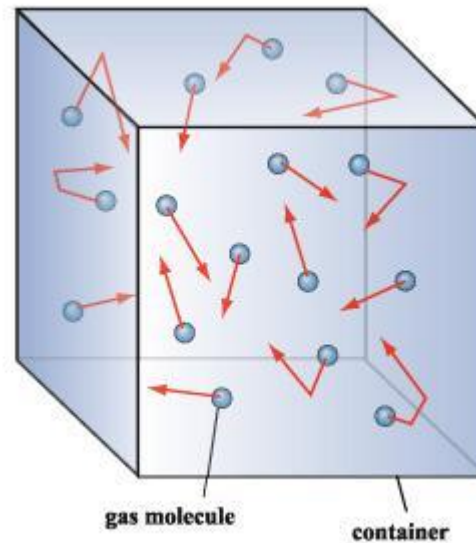
$$\text{Acceleration } \vec{a} = \frac{d\vec{v}}{dt}$$

Acceleration is determined by the force $\vec{F}$

$$\vec{F} = m\vec{a} = m \frac{d\vec{v}}{dt} = \frac{d\vec{p}}{dt}$$

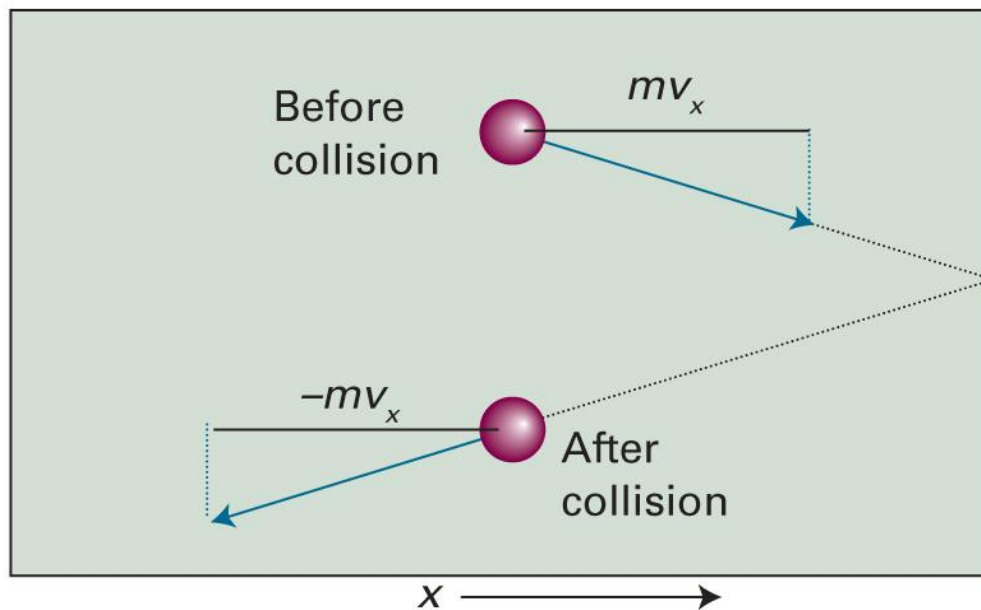
If $\vec{F} = 0$, $\frac{d\vec{p}}{dt} = 0$ (momentum conservation)

Derivation of Pressure



- Pressure $p = F/A$
- Force $\vec{F} = d\vec{p}/dt$: from collisions against the wall

The Change in Momentum



Change in momentum: $mv_x \rightarrow -mv_x$

$$\Delta p_{\text{particle}} = -mv_x - mv_x = -2mv_x$$

Momentum Change of the Wall

- The particle has the momentum change of

$$\Delta p_{\text{particle}} = -mv_x - mv_x = -2mv_x$$

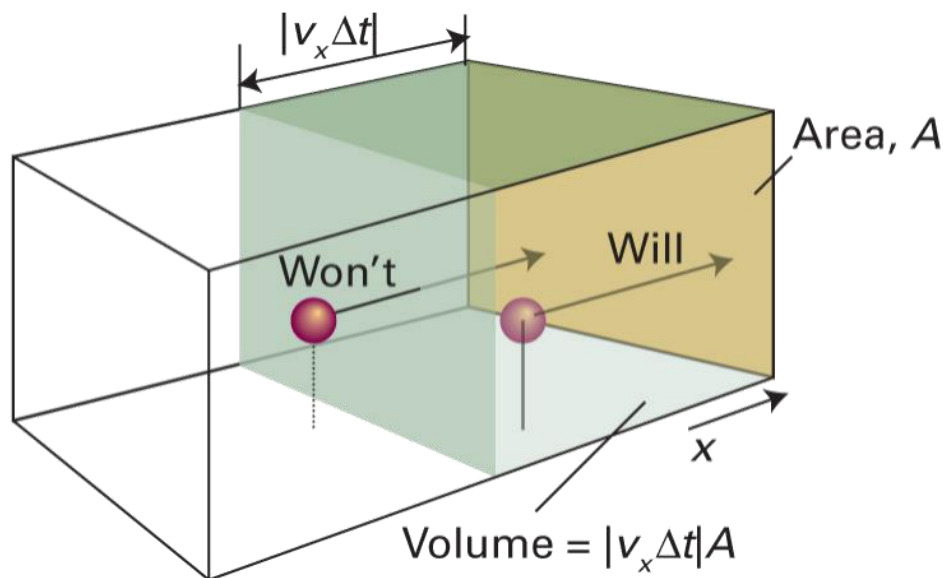
- The wall has the momentum change of

$$\Delta p_{\text{wall}} = 2mv_x$$

How Many Molecules Collide

Assume that all molecules have the same velocity.

For time Δt ,



Number of molecules = volume $\times$ density

$$= (A v_x \Delta t) \times (N_A n / V)$$

(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figure 1B.2)

Total Momentum Change

- For each molecule colliding the wall, the wall obtains

$$\Delta p_{\text{wall}} = 2mv_x$$

- Number of molecules = $(Av_x\Delta t) \times (N_A n/V)$
- 50% moving to the right, 50% to the left
- Total momentum change

$$\Delta p_{\text{wall, total}} = (2mv_x) \times (Av_x\Delta t) \left(\frac{N_A n}{V} \right) \times \frac{1}{2} = \frac{\overbrace{nmN_A}^M Av_x^2 \Delta t}{V}$$

Force and Pressure

- Total momentum change $\Delta p_{\text{wall,total}} = nMAv_x^2 \Delta t / V$
- Force exerted on the wall

$$F = \frac{\Delta p_{\text{wall,total}}}{\Delta t} = \frac{nMAv_x^2}{V}$$

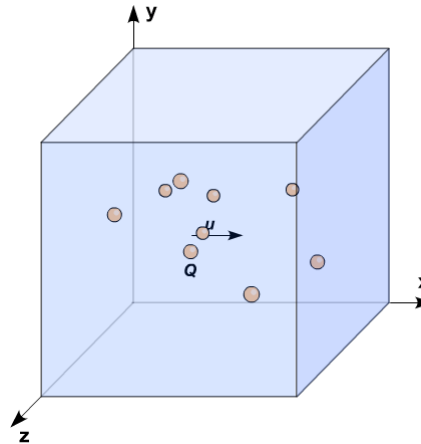
- Pressure exerted on the wall

$$p = \frac{F}{A} = \frac{nMv_x^2}{V}$$

(under the same velocity assumption)

Same Velocity Assumption

- In reality, each particle has a different velocity



- Replace the value by the mean value:

$$v_x^2 \rightarrow \langle v_x^2 \rangle$$

Pressure

$$p = \frac{nMv_x^2}{V} \rightarrow p = \frac{nM\langle v_x^2 \rangle}{V}$$

- Total velocity $v^2 = v_x^2 + v_y^2 + v_z^2$

$$\begin{aligned}\langle v^2 \rangle &= \langle v_x^2 \rangle + \langle v_y^2 \rangle + \langle v_z^2 \rangle \\ &= \langle v_x^2 \rangle + \langle v_x^2 \rangle + \langle v_x^2 \rangle = 3\langle v_x^2 \rangle\end{aligned}$$

- Final formula

$$p = \frac{nM\langle v^2 \rangle}{3V}$$

p - T Relationship

$$pV = \frac{1}{3} nM \langle v^2 \rangle$$

From the Maxwell-Boltzmann distribution,

$$\langle v^2 \rangle = v_{\text{rms}}^2 = \frac{3RT}{M}$$

Hence,

$$pV = nRT$$